

Table 1: Verification results for the autopilot properties derived from requirement φ_3 . The table reports whether each property is satisfied ($\checkmark$), violated ($\times$), or vacuously satisfied (v) for state machines learned from the **Descend** learning set. Colours indicate the input abstractions and requirement states: **Low**, **Med**, and **High** denote low, medium, and high input ranges, respectively, while **red**, **blue**, and **green** denote states in which requirement φ_3 is unsatisfied, satisfied, and robustly satisfied, respectively.

Property	Result
With the PitchWheel input	
From any state in which requirement φ_3 is unsatisfied , continuously Low PitchWheel input shall eventually cause the system to satisfy requirement φ_3 .	$\checkmark$
From any state in which requirement φ_3 is unsatisfied , continuously Med PitchWheel input shall eventually cause the system to satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is unsatisfied , continuously High PitchWheel input shall eventually cause the system to satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is satisfied , continuously Low PitchWheel input shall eventually cause the system to robustly satisfy requirement φ_3 .	$\checkmark$
From any state in which requirement φ_3 is satisfied , continuously Med PitchWheel input shall eventually cause the system to robustly satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is satisfied , continuously High PitchWheel input shall eventually cause the system to robustly satisfy requirement φ_3 .	$\times$
With the Throttle input	
From any state in which requirement φ_3 is unsatisfied , continuously Low Throttle input shall eventually cause the system to satisfy requirement φ_3 .	$\checkmark$
From any state in which requirement φ_3 is unsatisfied , continuously High Throttle input shall eventually cause the system to satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is satisfied , continuously Low Throttle input shall eventually cause the system to robustly satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is satisfied , continuously High Throttle input shall eventually cause the system to robustly satisfy requirement φ_3 .	$\times$
With both PitchWheel and Throttle inputs	
From any state in which requirement φ_3 is unsatisfied , continuously Low PitchWheel and Low Throttle inputs shall eventually cause the system to satisfy requirement φ_3 .	$\checkmark$
From any state in which requirement φ_3 is unsatisfied , continuously Low PitchWheel and High Throttle inputs shall eventually cause the system to satisfy requirement φ_3 .	v
From any state in which requirement φ_3 is unsatisfied , continuously Med PitchWheel and Low Throttle inputs shall eventually cause the system to satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is unsatisfied , continuously Med PitchWheel and High Throttle inputs shall eventually cause the system to satisfy requirement φ_3 .	v
From any state in which requirement φ_3 is unsatisfied , continuously High PitchWheel and Low Throttle inputs shall eventually cause the system to satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is unsatisfied , continuously High PitchWheel and High Throttle inputs shall eventually cause the system to satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is satisfied , continuously Low PitchWheel and Low Throttle inputs shall eventually cause the system to robustly satisfy requirement φ_3 .	$\checkmark$
From any state in which requirement φ_3 is satisfied , continuously Low PitchWheel and High Throttle inputs shall eventually cause the system to robustly satisfy requirement φ_3 .	$\checkmark$
From any state in which requirement φ_3 is satisfied , continuously Med PitchWheel and Low Throttle inputs shall eventually cause the system to robustly satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is satisfied , continuously Med PitchWheel and High Throttle inputs shall eventually cause the system to robustly satisfy requirement φ_3 .	$\times$
From any state in which requirement φ_3 is satisfied , continuously High PitchWheel and Low Throttle inputs shall eventually cause the system to robustly satisfy requirement φ_3 .	v
From any state in which requirement φ_3 is satisfied , continuously High PitchWheel and High Throttle inputs shall eventually cause the system to robustly satisfy requirement φ_3 .	$\times$